8 Fig. 8.1 shows a rectangular coil. The coil has 25 turns with dimensions of 15.0 cm by 6.0 cm.

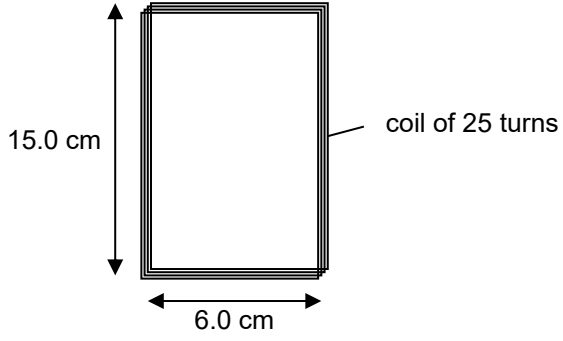

Fig. 8.1

(a) **(i)** Define magnetic flux.

……...

…….. [1]

(ii) A uniform magnetic field of flux density 25 mT is at right angles to the plane of the coil initially.

Calculate the magnetic flux through the coil at this instance.

magnetic flux =…….....Wb [1]

(b) **(i)** Fig. 8.2 shows the sinusoidal variation with time *t* of the magnetic flux density *B* that is passing through the rectangular coil in Fig. 8.1.

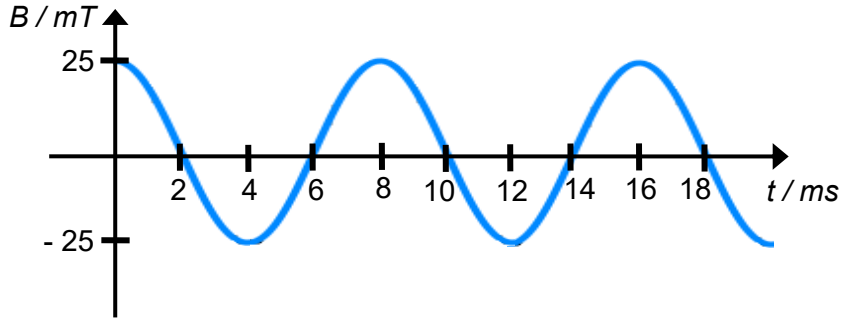

Fig. 8.2

Using Faraday's Law, explain why the variation in magnetic flux density passing through the coil as shown in Fig. 8.2 leads to a generation of sinusoidal alternating e.m.f.

……

……..

……..

…….. [3]

(ii) Show that the equation for the induced e.m.f. of the coil is

$$E = NAB_o \frac{2\pi}{T} \sin\left(\frac{2\pi}{T} t\right)$$

where *N* is the number of turns in the coil, *A* is the area of coil, B_0 is the maximum magnetic flux density and *T* is the period of the change in magnetic flux density.

[2]

(iii) Hence, determine the maximum magnitude of the induced e.m.f. of the coil.

maximum e.m.f. = ……. V [2]

(iv) On Fig. 8.3, sketch a graph to show the variation with time of the e.m.f. induced in the coil from 0 ms to 16 ms. [2]

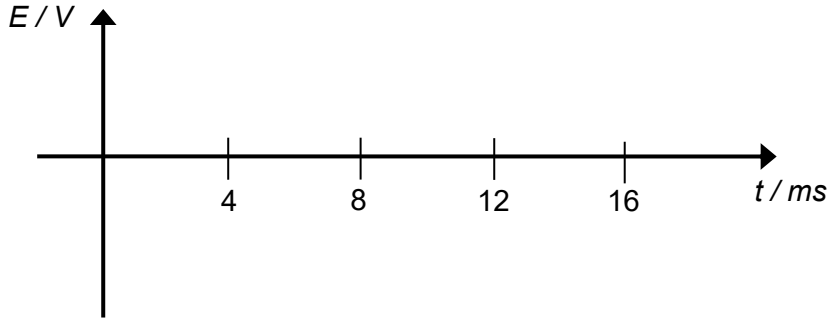

Fig. 8.3

(c) **(i)** The AC generated by the rectangular coil passes through a transformer. The primary coil of the ideal transformer has 15 turns.

Assuming that the transformer is ideal, calculate the number of turns in the secondary coil if the value of root-mean-square potential difference of the output at the secondary coil is 240 V.

number of turns =……..... [3]

(ii) A non-ideal transformer has an input e.m.f. of 12 V while the secondary coil draws a current of 0.25 A with a potential difference of 240 V.

If the efficiency of the transformer is 81 %, determine the current in the primary coil.

(d) In many distribution systems for electrical energy, the power is transmitted using alternating current at high voltages.

Suggest and explain an advantage, one in each case, for the use of

(i) alternating voltages,

.....

.....

..... [2]

(ii) high voltages.

.....

.....

..... [2]